IGCSE 0607 Practice Questions — E6.1 — Combined Questions & Markscheme

Units of measure

Total Questions: 18 — **Total Marks:** 129 Marks

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- **E6.1** Units of measure
  → **Paper 2 (Non-Calculator)** *(7 Qs)* · **Paper 4 (Calculator)** *(11 Qs)*

1 **(a)** Change 52000 cm^2 into m^2.

..... m^2 **[1]**

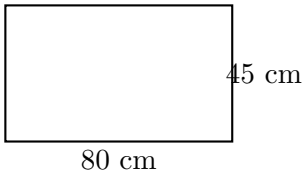
(b) Change 4.7 m^3 into litres.

..... litres **[1]**

(c) Change 850 millilitres into cm^3.

..... cm^3 **[1]**

2 A rectangular piece of card and a water jug are being measured.



(a) Calculate the area of the rectangular piece of card in square metres (m^2).

..... m^2 **[2]**

(b) The jug holds 3.2 litres of water.
Write down this volume in cm^3.

..... cm^3 **[1]**

3 **(a)** Change 3.5 m^2 into cm^2.

..... cm^2 **[1]**

(b) A rectangular tank measures 60 cm by 45 cm by 30 cm.
Calculate the volume of the tank in litres.

..... litres **[3]**

4 Complete each of the following metric unit conversions.

(a) Convert 4.75 km into metres.

..... m **[1]**

(b) Convert 320 g into kilograms.

..... kg **[1]**

(c) Convert 2.4 m^2 into cm^2.

..... cm^2 **[1]**

(d) Convert 6500 cm^3 into litres.

..... litres **[1]**

5 A solid wooden block is in the shape of a cuboid.
The dimensions of the block are 40 cm, 250 mm and 0.2 m.

(a) Work out the volume of the block in cm^3.

..... cm^3 **[2]**

(b) Convert your answer to part (a) into litres.

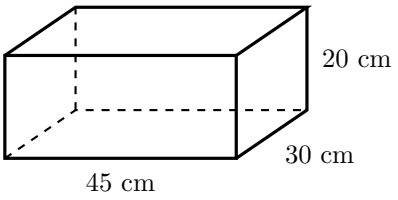
..... litres **[1]**

(c) The density of the wood is 0.6 g/cm^3.
Calculate the mass of the block in kilograms.

..... kg **[2]**

6 The diagram shows a closed rectangular box of length 45 cm, width 30 cm, and height 20 cm.

NOT TO SCALE



(a) Calculate

(i) the total surface area of the box in cm^2,

..... cm^2 **[2]**

(ii) your answer to part (a)(i) converted into m^2.

..... m^2 **[1]**

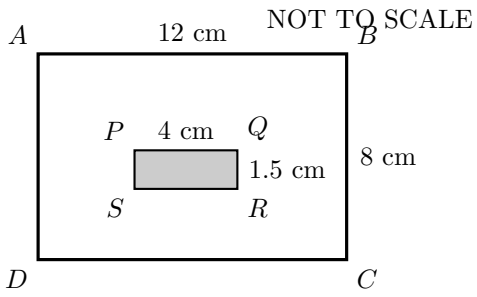
(b) Calculate the volume of the box in cm^3, and write down its capacity in litres.

Volume = ... , Capacity = ... **[2]**

(c) The empty box has a mass of 650 g. It is packed with 18 identical books, each of mass 1.15 kg. Calculate the total mass of the packed box in kilograms.

..... kg **[2]**

7 The diagram shows the scale plan of a rectangular park $ABCD$ containing a rectangular pond $PQRS$.
The plan is drawn to a scale of $1 : 25\,000$.
On the plan, $AB = 12$ cm, $BC = 8$ cm, $PQ = 4$ cm, and $QR = 1.5$ cm.



(a) Calculate the actual perimeter of the park $ABCD$, giving your answer in kilometres.

..... km **[2]**

(b) Calculate

(i) the area of the park excluding the pond, in cm^2, on the plan,

..... cm^2 **[2]**

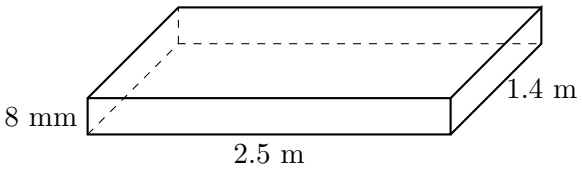
(ii) the actual area of the park excluding the pond, giving your answer in km^2.

..... km^2 **[2]**

(c) The actual pond $PQRS$ has a uniform depth of 1.6 m. Calculate the capacity of the pond when it is completely full, giving your answer in litres in standard form.

..... litres **[3]**

8 A rectangular block of metal has length 2.5 m, width 1.4 m and thickness 8 mm.



NOT TO SCALE

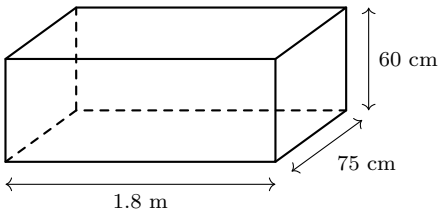
(a) Work out the volume of the block of metal in cm^3.

..... cm^3 **[2]**

(b) The mass of 1 cm^3 of this metal is 7.8 g.
Work out the mass of the block of metal in kilograms.

..... kg **[2]**

9 The diagram shows a rectangular water tank with internal length 1.8 m, internal width 75 cm, and internal height 60 cm.



(a) Calculate the total internal capacity of the tank in litres.

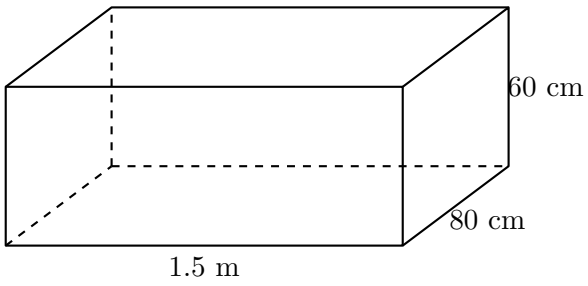
Capacity = ... litres **[3]**

(b) The empty tank is filled with water from a tap flowing at a rate of 450 ml per second.

Calculate the time taken to fill the tank completely.
Give your answer in minutes.

Time = ... minutes **[3]**

10 A metal water tank is made in the shape of a cuboid.



(a) Calculate the capacity of the water tank in litres.

..... litres **[3]**

(b) A solid metal sphere is melted down to make the tank.
The metal sphere has a volume of 0.024 m^3.
The mass of 1 cm^3 of this metal is 7.8 g.

Calculate the total mass of the sphere, giving your answer in kilograms.

..... kg **[3]**

11 During a storm, 15 mm of rain falls on a flat rectangular roof measuring 15 m by 12 m.

(a) Calculate the volume of rain that falls on the roof.
Give your answer in litres.

..... litres **[3]**

(b) The water from the roof flows into a cylindrical tank with a base radius of 60 cm.
Calculate the increase in the depth of the water in the tank.
Give your answer in centimetres, correct to 1 decimal place.

..... cm **[4]**

12 A solid metal sphere has a radius of 15 cm.
The volume, V, of a sphere with radius r is $V = \frac{4}{3}\pi r^3$.

(a) Calculate the volume of the sphere in m^3.
Give your answer in standard form.

..... m^3 **[3]**

(b) The mass of the sphere is 125 kg.
Calculate the density of the metal in g/cm^3.

..... g/cm^3 **[3]**

(c) The sphere is melted down to make completely solid identical cones.
Each cone has a base radius of 24 mm and a height of 5.5 cm.
The volume, V, of a cone with radius r and height h is $V = \frac{1}{3}\pi r^2 h$.

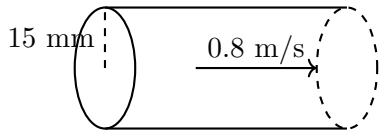
(i) Work out the volume of one cone in cm^3.

..... cm^3 **[3]**

(ii) Work out the maximum number of completely solid cones that can be made.

..... **[2]**

13 Water flows through a cylindrical pipe into an empty rectangular pool.
The pool measures 4 m by 2.5 m.



(a) The cylindrical pipe has a radius of 15 mm.
Water flows through the pipe at a speed of 0.8 m/s.

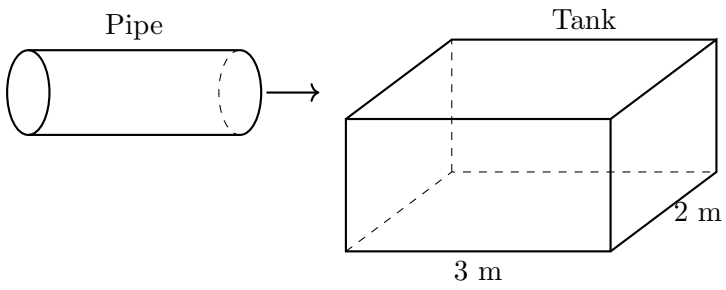
Show that the volume of water flowing out of the pipe in one second is 0.565 litres, correct to 3 significant figures.

[3]

(b) Calculate the time taken for the depth of water in the pool to reach 1.2 m.
Give your answer in hours and minutes, correct to the nearest minute.

..... hours ..... minutes **[4]**

14 Water flows through a cylindrical pipe of internal radius 2.5 cm at a speed of 5.4 km/h.
The water flows into an empty rectangular tank of length 3 m and width 2 m.



NOT TO SCALE

(a) Show that the volume of water flowing into the tank in 1 minute is 176.7 litres, correct to 1 decimal place.

[3]

(b) Calculate the depth of the water in the tank after 45 minutes.
Give your answer in centimetres.

..... cm **[4]**

15 A farmer owns a rectangular field measuring 450 m by 320 m.

(a) Calculate the area of the field in km^2.

Area = ... km^2 **[2]**

(b) A liquid fertiliser is sprayed over the entire field at a rate of 25 ml per m^2.
The fertiliser is sold in drums each containing 0.18 m^3 of liquid.

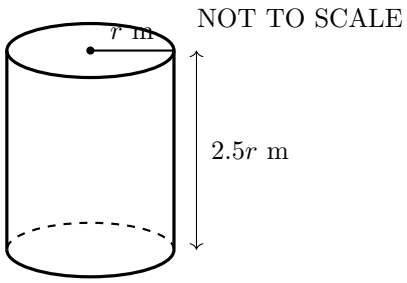
(i) Calculate the total volume of fertiliser required for the field, in m^3.

Volume = ... m^3 **[3]**

(ii) Each drum of fertiliser has a mass of 216 kg when full, and the empty drum has a mass of 9 kg.
Calculate the mass, in grams, of 1 cm^3 of the liquid fertiliser.

Mass = ... g **[3]**

16 A closed cylindrical water tank has internal radius r metres and internal height h metres, where $h = 2.5r$.
The internal capacity of the tank is 15 700 litres.



(a) Convert 15 700 litres into m^3, and hence calculate the value of r, giving your answer correct to 3 significant figures.

Volume = ... , r = ... **[3]**

(b) The entire inside surface of the closed tank is coated with a protective waterproof layer of uniform thickness 1.5 mm. Using $r = 1.26$, estimate the volume of waterproof coating used by multiplying the internal surface area by the thickness.
Give your answer in cm^3 correct to 3 significant figures.

..... cm^3 **[3]**

(c) Water flows out of the full tank through a pipe at a constant rate of 450 ml per second.
Calculate the time taken to empty the 15 700 litres of water from the tank, giving your answer in hours and minutes correct to the nearest minute.

..... hours ..... minutes **[3]**

(d) The water in the tank has a density of 1.02 g per cm^3.
Calculate the total mass of the 15 700 litres of water, giving your answer in tonnes, where 1 tonne = 1000 kg.

..... tonnes **[2]**

17 Water flows through a cylindrical pipe of radius 2.5 cm at a speed of 18 km/h.



(a) Calculate the volume of water, in litres, that flows through the pipe in 1 minute.

..... litres **[5]**

(b) The water from the pipe is used to fill an empty rectangular reservoir measuring 50 m by 30 m by 4 m.
Calculate the time it takes to fill the reservoir completely.
Give your answer in days and hours, correct to the nearest hour.

days ..... , hours ..... **[7]**

18 Water flows through a cylindrical pipe of radius 2.5 cm at a speed of 1.6 m/s.

(a) Calculate the volume of water, in litres, that flows through the pipe in one minute.
Give your answer correct to 3 significant figures.

..... litres **[4]**

(b) This water flows into an empty rectangular tank measuring 4.2 m by 3.5 m by 2.4 m.
Calculate the time taken to fill the tank completely.
Give your answer in hours and minutes, correct to the nearest minute.

hours ..... , minutes ..... **[5]**

(c) The outside vertical walls of the rectangular tank are to be painted.
1 litre of paint covers 45000 cm^2.
Paint is sold in 2.5-litre tins.
Work out the minimum number of tins of paint needed.

..... tins **[6]**

1 **(a)** 5.2 **[B1]**

(b) 4700 **[B1]**

(c) 850 **[B1]**

[Back to Question] *[Total Marks: 3]*

2 **(a)** $80 \times 45 = 3600$ **[M1]**

0.36 **[A1]**

(b) 3200 **[B1]**

[Back to Question] *[Total Marks: 3]*

3 **(a)** 35000 **[B1]**

(b) $60 \times 45 \times 30$ OR 2700×30 OR 81000 **[M1]**

their $81000 \div 1000$ **[M1]**

81 **[A1]**

[Back to Question] *[Total Marks: 4]*

4 **(a)** 4750 **[B1]**

(b) 0.32 **[B1]**

(c) 24 000 **[B1]**

(d) 6.5 **[B1]**

[Back to Question] *[Total Marks: 4]*

5 **(a)** Converts units correctly to cm: 250 mm = 25 cm AND 0.2 m = 20 cm **[B1]**

20000 **[A1]**

(b) 20 **[B1]**

(c) Uses Mass = Density × Volume: 0.6×20000 (= 12000 g) **[M1]**

12 **[A1]**

[Back to Question] *[Total Marks: 5]*

6 **(a)** **(i)** $2(45 \times 30 + 45 \times 20 + 30 \times 20) = 2(1350 + 900 + 600)$ **[M1]**

5700 **[A1]**

(ii) 0.57 **[B1]**

(b) Volume $= 45 \times 30 \times 20 = 27\,000\ cm^3$ **[B1]**

Capacity = 27 litres **[B1]**

(c) 650 g = 0.65 kg and $18 \times 1.15 = 20.7$ kg **[M1]**

21.35 **[A1]**

[Back to Question] *[Total Marks: 7]*

7 **(a)** Perimeter on plan $= 2(12 + 8) = 40$ cm and actual perimeter $= 40 \times 25\,000 = 1\,000\,000$ cm **[M1]**

10 **[A1]**

(b) **(i)** $(12 \times 8) - (4 \times 1.5) = 96 - 6$ **[M1]**

90 **[A1]**

(ii) 1 cm represents 0.25 km, so 1 cm^2 represents $0.25^2 = 0.0625$ km^2 (or $90 \times 25\,000^2 = 5.625 \times 10^{10}$ cm^2) **[M1]**

5.625 **[A1]**

(c) Actual pond dimensions: $4 \times 250 = 1000$ m and $1.5 \times 250 = 375$ m **[M1]**

Volume $= 1000 \times 375 \times 1.6 = 600\,000$ m^3 **[M1]**

6×10^8 **[A1]**

[Back to Question] *[Total Marks: 9]*

8 **(a)** Converts all dimensions to cm: 250, 140, 0.8 **[M1]**

$250 \times 140 \times 0.8 = 28000$ **[A1]**

(b) Multiplies volume by 7.8: $28000 \times 7.8 = 218400$ **[M1]**

218.4 **[A1]**

[Back to Question] *[Total Marks: 4]*

9 **(a)** Converts dimensions consistently, e.g. 1.8 m = 180 cm (or 0.75 m and 0.6 m) **[B1]**

Calculates volume: $180 \times 75 \times 60 = 810\,000$ cm^3 (or 0.81 m^3) **[M1]**

810 **[A1]**

(b) Converts 810 litres = 810 000 ml (or 450 ml/s = 0.45 l/s) **[M1]**

Calculates time in seconds: $\frac{810\,000}{450} = 1800$ seconds **[M1]**

30 **[A1]**

[Back to Question] *[Total Marks: 6]*

10 **(a)** Converts dimensions to consistent units, e.g. 150 cm, 80 cm, 60 cm OR 1.5 m, 0.8 m, 0.6 m **[M1]**

$150 \times 80 \times 60 = 720000$ OR $1.5 \times 0.8 \times 0.6 = 0.72$ **[M1]**

720 **[A1]**

(b) $0.024 \times 1000000 = 24000$ **[M1]**

Mass $= 24000 \times 7.8 = 187200$ **[M1]**

187.2 **[A1]**

[Back to Question] *[Total Marks: 6]*

11 **(a)** Conversion of units: 15 $mm = 0.015$ m OR 15 $m = 1500$ cm etc. **[M1]**

Volume $= 15 \times 12 \times 0.015$ (or eq) **[M1]**

2700 **[A1]**

(b) Use of consistent units for tank volume equation (e.g. $r = 0.6$ m and $V = 2.7$ m^3 OR $V = 2700000$ cm^3 and $r = 60$ cm) **[M1]**

$\pi \times 0.6^2 \times h = 2.7$ OR $\pi \times 60^2 \times h = 2700000$ **[M1]**

$h = \dfrac{2.7}{0.36\pi}$ (m) OR $h = \dfrac{2700000}{3600\pi}$ (cm) **[M1]**

238.7 **[A1]**

[Back to Question] *[Total Marks: 7]*

12 **(a)** $\dfrac{4}{3} \times \pi \times 15^3$ OR 4500π OR 14137 **[M1]**

their 14137 $\div$ 1000000 OR $\dfrac{4}{3} \times \pi \times 0.15^3$ **[M1]**

1.41×10^{-2} **[A1]**

(b) 125 kg = 125000 g **[B1]**

$\dfrac{125000}{4500\pi}$ OR $\dfrac{125000}{14137}$ **[M1]**

8.84 (8.841 . . .) **[A1]**

(c) **(i)** 24 mm = 2.4 cm **[B1]**

$\dfrac{1}{3} \times \pi \times 2.4^2 \times 5.5$ **[M1]**

33.2 (33.17 . . .) OR 10.56π **[A1]**

(ii) $\dfrac{4500\pi}{\text{their } 33.17}$ OR 426.1 . . . **[M1]**

426 **[A1]**

[Back to Question] *[Total Marks: 11]*

13 **(a)** Identifies consistent units for radius and length (e.g. $r = 1.5$ cm and length = 80 cm) **[M1]**

Volume in one second $= \pi \times 1.5^2 \times 80 = 180\pi$ cm^3 $(= 565.48 \ldots$ $cm^3)$ **[M1]**

Converts to litres $(180\pi \div 1000)$ and confirms $0.56548 \cdots \approx 0.565$ **[A1]**

(b) Volume of pool required $= 4 \times 2.5 \times 1.2 = 12$ m^3 OR 12000 litres **[M1]**

Time in seconds $= 12000 \div 0.56548 \ldots$ OR $12000 \div \dfrac{180\pi}{1000}$ **[M1]**

$= 21220.6\ldots$ seconds $(353.6\ldots$ minutes) **[A1]**

5 hours 54 minutes **[A1]**

[Back to Question] *[Total Marks: 7]*

14 **(a)** Converts speed to cm/min: $5.4 \times 1000 \times 100 \div 60 = 9000$ **[M1]**

Calculates volume in 1 minute: $\pi \times 2.5^2 \times 9000 = 56250\pi\ cm^3$ **[M1]**

Converts cm^3 to litres and evaluates: $56250\pi \div 1000 = 176.714\ldots$ **[A1]**

(b) Calculates total volume in 45 minutes: $56250\pi \times 45 = 2531250\pi\ cm^3$ (or uses litres: $176.714\cdots \times 45 = 7952.1\ldots$ litres) **[M1]**

Converts tank base dimensions to cm and finds area: $300 \times 200 = 60000\ cm^2$ **[M1]**

Sets up equation for depth: $\text{Depth} = \dfrac{2531250\pi}{60000}$ **[M1]**

133 (or 132.5 to 132.6) **[A1]**

[Back to Question] *[Total Marks: 7]*

15 **(a)** Calculates area in m^2: $450 \times 320 = 144\,000\ m^2$ (or 0.45×0.32) **[M1]**

0.144 **[A1]**

(b) **(i)** Total volume in ml $= 144\,000 \times 25 = 3\,600\,000$ ml **[M1]**

Converts ml to litres or cm^3: 3600 litres or $3\,600\,000\ cm^3$ **[M1]**

3.6 **[A1]**

(ii) Mass of liquid in one drum $= 216 - 9 = 207$ kg $= 207\,000$ g **[B1]**

Volume of one drum $= 0.18\ m^3 = 180\,000\ cm^3$ **[B1]**

$\dfrac{207\,000}{180\,000} = 1.15$ **[A1]**

[Back to Question] *[Total Marks: 8]*

16 **(a)** Volume $= 15.7\ m^3$ **[B1]**

$\pi r^2(2.5r) = 15.7$ giving $r^3 = \dfrac{15.7}{2.5\pi}$ $(= 1.99898\ldots)$ **[M1]**

$r = 1.26$ **[A1]**

(b) Internal surface area $= 2\pi(1.26)^2 + 2\pi(1.26)(3.15) = 34.913\ldots\ m^2$ $(= 349\,131\ldots\ cm^2)$ **[M1]**

Volume $= 349\,131\cdots \times 0.15$ (or $34.913\cdots \times 0.0015 \times 10^6$) **[M1]**

52 400 **[A1]**

(c) 15 700 litres $= 15\,700\,000$ ml and time $= \dfrac{15\,700\,000}{450} = 34\,888.88\ldots$ seconds **[M1]**

$\dfrac{34\,888.88\ldots}{60} = 581.48\ldots$ minutes **[M1]**

9 hours 41 minutes **[A1]**

(d) Mass $= 15\,700\,000 \times 1.02 = 16\,014\,000$ g $(= 16\,014$ kg) **[M1]**

16.014 **[A1]**

[Back to Question] *[Total Marks: 11]*

17 **(a)** Cross-sectional area $= \pi \times 2.5^2$ $(= 6.25\pi)$ **[M1]**

Speed conversion: $18\ km/h = \dfrac{18 \times 1000 \times 100}{60}\ cm/min$ $(= 30000\ cm/min)$ **[M1]**

Volume per minute $= 6.25\pi \times 30000$ $(= 187500\pi\ cm^3)$ **[M1]**

Convert to litres: $\dfrac{187500\pi}{1000}$ **[M1]**

589 or 589.0 to 589.1 (187.5π) **[A1]**

(b) Capacity of reservoir $= 50 \times 30 \times 4 = 6000\ m^3$ OR 6000000 litres **[M1]**

Time in minutes $= \dfrac{6000000}{187.5\pi}$ (or 6000000÷ their (a)) $(= 10185.9...)$ **[M1]**

Time in hours $= \dfrac{10185.9...}{60}$ $(= 169.76...)$ **[M1]**

$169.76... \div 24 = 7.07...$ days so 7 full days **[M1]**

$0.07... \times 24 = 1.76...$ hours (or $169.76... - 7 \times 24$) **[M1]**

7 **[A1]**

2 **[A1]**

[Back to Question] *[Total Marks: 12]*

18 **(a)** $\pi \times 2.5^2 \times 160$ OR $\pi \times 0.025^2 \times 1.6$ **[M1]**

$\times 60$ **[M1]**

$\div 1000$ (if using cm) OR $\times 1000$ (if using m) **[M1]**

188 $(188.4\ldots)$ **[A1]**

(b) Volume of tank $= 4.2 \times 3.5 \times 2.4 = 35.28\ m^3$ **[M1]**

$35.28 \times 1000 = 35280$ litres **[M1]**

Time $= \dfrac{35280}{\text{their } 188.49}$ OR $\dfrac{35280}{60\pi}$ **[M1]**

187 minutes $(187.1\ldots)$ **[A1]**

3 hours 7 minutes **[A1]**

(c) Calculates lateral area correctly: $2 \times 4.2 \times 2.4 + 2 \times 3.5 \times 2.4$ OR $36.96\ m^2$ [M2]
(If incorrect, M1 for one correct face area e.g., 4.2×2.4)

$36.96 \times 10000 = 369600\ cm^2$ OR $45000 \div 10000 = 4.5\ m^2$ **[M1]**

Litres of paint required $= \dfrac{369600}{45000}$ OR $\dfrac{36.96}{4.5}$ **[M1]**

Number of tins $= \dfrac{8.21\ldots}{2.5}$ $(= 3.28\ldots)$ **[M1]**

4 **[A1]**

[Back to Question] *[Total Marks: 15]*